NR2DT-04: Step 4 — Translate

> **Series:** NR2DT — New Relic to Dynatrace Migration Steps | **Notebook:** 4 of 10 | **Created:** April 2026 | **Last Updated:** 10/31/2026

Overview

Goal of this step: convert every NRQL query in the inventory to DQL. Auto-translate HIGH; review MEDIUM; hand-translate LOW. Resolve every gap before any artifact migrates.

Procedural — see **NRLC-02** for the compiler architecture, the 292 patterns, and confidence theory.

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step

Requirement	Details
Completed	NR2DT-03 — Design
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	NRLC-02 (NRQL→DQL Translation deep dive)

1. What You'll Produce

Artifact	Format	Purpose
`translated-queries.csv`	CSV	All queries with final DQL + confidence
`manual-translations.md`	Markdown	LOW-confidence hand-translations with rationale
`dql-validation-report.md`	Markdown	Tier 1 syntax validation results
`translation-coverage.md`	Markdown	% of queries translated; remaining gaps

2. Run the Translator

```
# Re-run translation against the latest inventory (Step 1 produced inventory/all-nrql.csv)
python3 migrate.py batch --file inventory/all-nrql.csv --output translated-queries.csv
```

Output columns — exactly four: `nrql`, `dql`, `confidence` (HIGH / MEDIUM / LOW, or `FAILED` when the query did not compile, in which case the `dql` cell holds the error), `warnings`. Use `batch` for the CSV: `compile` has no `--report` option (it accepts only `--interactive`, `--file`, `--output` and `--validate`) and writes DQL text, not a CSV.

> **Sources:** migrate.py @ 78cbfce (tool repo, GitHub), read 10/05/2026.

Sort by confidence ascending — attack LOW first, then MEDIUM, save HIGH for sample-validate.

3. Review MEDIUM Confidence

MEDIUM means the translation is structurally correct but has edge cases worth verifying. Common review checklist:

Pattern	What to verify
`COMPARE WITH`	Time alignment correct; append produces the expected shape
Subquery via `lookup`	Lookup target exists as queryable Grail data
`SLIDE BY`	The compiler emits `rolling()`, which does not exist in DQL; rewrite by hand per NRLC-02
Nested `if()` in `FACET CASES`	Nested DQL emits the right structure
Custom event types	Source mapping (FROM → fetch) correct

Process: mark each MEDIUM as APPROVED or NEEDS-WORK in the CSV. NEEDS-WORK becomes a hand-translation.

4. Hand-Translate LOW Confidence

LOW translations contain TODO markers or are missing entirely. Common causes:

Cause	Action
`apdex(field, t:N)`	Configure DT Apdex via custom DQL or use Dynatrace Intelligence SLO
`funnel(...)`	Reformulate as nested `append` / `lookup` chain
`cohort(...)`	No DT equivalent — reformulate the requirement
Custom event type with no source mapping	Add ingest mapping via OpenPipeline first

Document each hand-translation in `manual-translations.md` with the source NRQL, the new DQL, and the rationale.

If a query can't be translated and the requirement can't be reformulated, document it as **deferred** — it joins the gap analysis from Step 1.

5. Validate Translated DQL

Tier 1 syntax validation — confirm every final DQL query parses against the target tenant.

The tool helps only partly. `compile --file` takes the **NRQL** list (one query per line), not the CSV of translated DQL — fed the CSV, it compiles the header and every DQL row as if they were NRQL:

```
python3 migrate.py compile --file inventory/all-nrql.txt --output
translated.dql
```

It prints a succeeded / failed count and writes each query as `-- Original: <nrql>` followed by its DQL (or `-- Error:`). It does **not** live-validate in file mode — `--validate` is honoured only for a single query — and there is no auto-fixer on this path. Check one query against the live tenant with:

```
python3 migrate.py compile --validate "SELECT count(*) FROM Transaction"
```

For the full set, run each final DQL query (including reviewed-MEDIUM and hand-translated ones from the CSV) in the target tenant — in a notebook, or with `dtctl verify query` — and record pass / fail in `dql-validation-report.md`.

> **Sources:** migrate.py @ 78cbfce (tool repo, GitHub), read 10/05/2026.

Tier 2 tenant validation (referenced metrics/entities/buckets exist) runs in Step 8 before each component imports.

6. Step Exit Criteria

S4 — Translation Complete

- [ ] Every query has a final DQL (auto, reviewed-MEDIUM, or hand-translated)
- [ ] All translations pass Tier 1 syntax validation
- [ ] LOW-confidence count is 0 (all triaged)
- [ ] `manual-translations.md` documents every LOW resolution
- [ ] Deferred queries documented with deferred-rationale

Next step: NR2DT-05 — Migrate Dashboards & Alerts.

This notebook was AI-generated from community-submitted and publicly available sources, including the open-source NewRelic-to-Dynatrace-Migration-Utilities, nrql-engine, and nrql-translator projects. This notebook series is not officially supported by Dynatrace or New Relic. Always verify information against the official Dynatrace documentation and New Relic documentation.